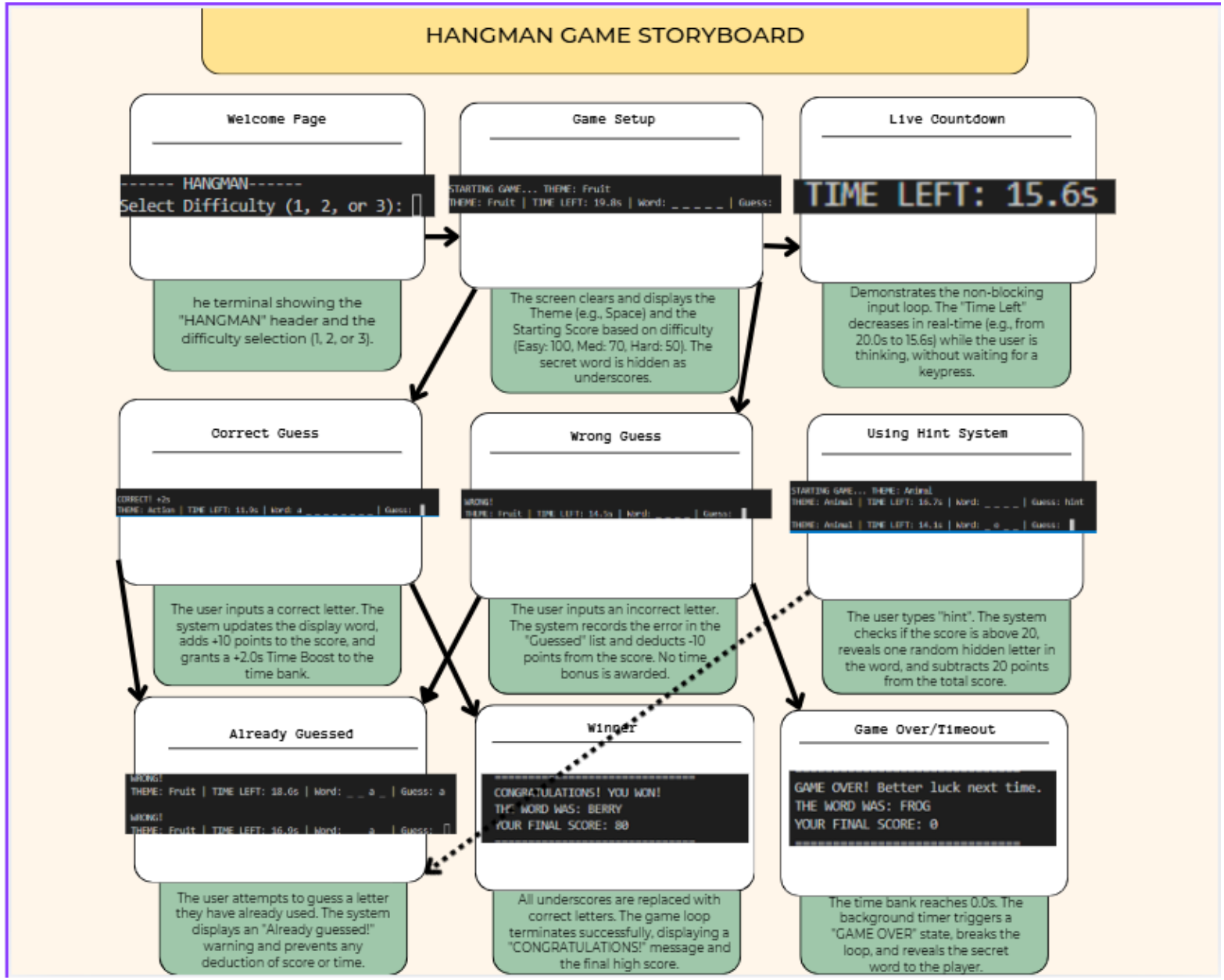
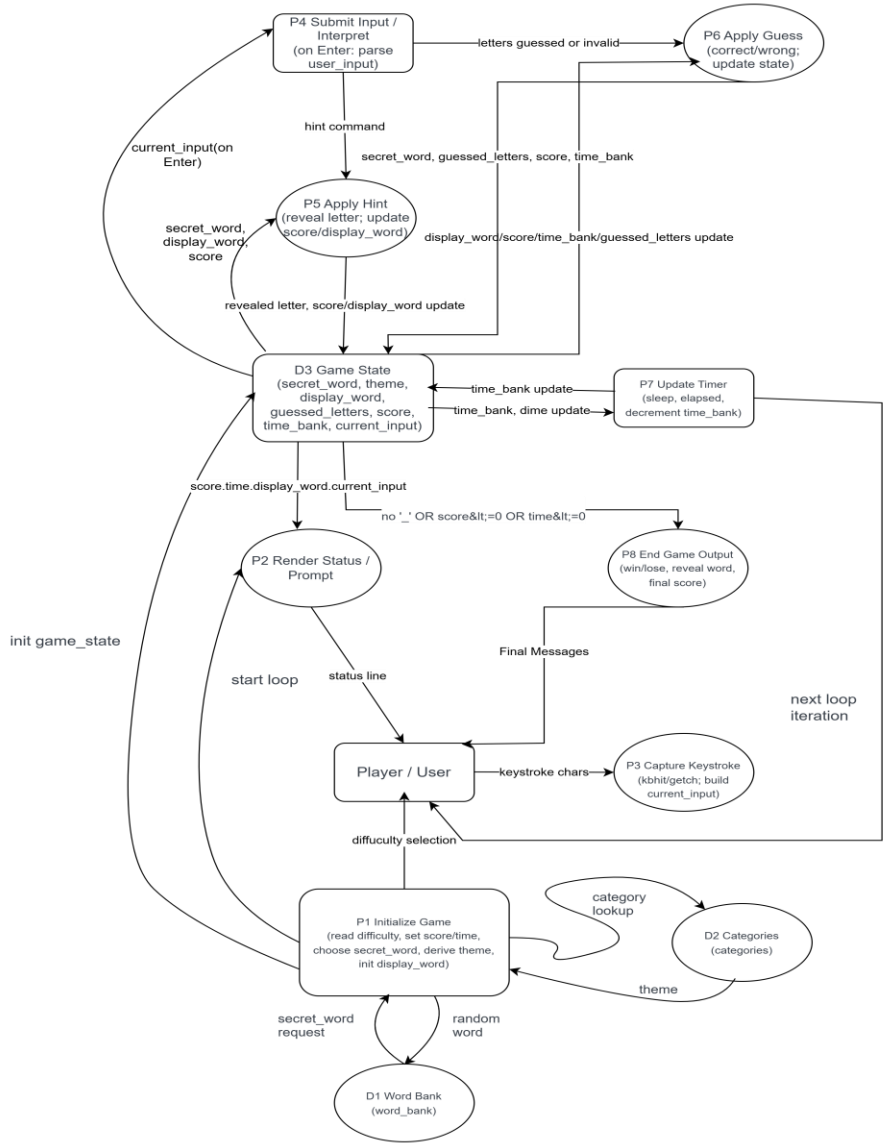
Software Engineering Documentation (Hangman)

Storyboard:


HANGMAN GAME STORYBOARD
Welcome Page
------ HANGMAN------
Select Difficulty (1, 2, or 3):
he terminal showing the "HANGMAN" header and the difficulty selection (1, 2, or 3).
Game Setup
STARTING GAME... THEME: Fruit
THEME: Fruit | TIME LEFT: 19.8s | Word: _ _ _ _ _ | Guess:
The screen clears and displays the Theme (e.g., Space) and the Starting Score based on difficulty (Easy: 100, Med: 70, Hard: 50). The secret word is hidden as underscores.
Live Countdown
TIME LEFT: 15.6s
Demonstrates the non-blocking input loop. The "Time Left" decreases in real-time (e.g., from 20.0s to 15.6s) while the user is thinking, without waiting for a keypress.
Correct Guess
CORRECT! +2s
THEME: Action | TIME LEFT: 11.9s | Word: a _ _ _ _ _ _ _ _ | Guess:
The user inputs a correct letter. The system updates the display word, adds +10 points to the score, and grants a +2.0s Time Boost to the time bank.
Wrong Guess
WRONG!
THEME: Fruit | TIME LEFT: 14.5s | Word: _ _ _ _ _ | Guess:
The user inputs an incorrect letter. The system records the error in the "Guessed" list and deducts -10 points from the score. No time bonus is awarded.
Using Hint System
STARTING GAME... THEME: Animal
THEME: Animal | TIME LEFT: 16.7s | Word: _ _ _ _ | Guess: hint
THEME: Animal | TIME LEFT: 14.1s | Word: _ e _ _ | Guess:
The user types "hint". The system checks if the score is above 20, reveals one random hidden letter in the word, and subtracts 20 points from the total score.
Already Guessed
WRONG!
THEME: Fruit | TIME LEFT: 18.6s | Word: _ _ a _ | Guess: a
WRONG!
THEME: Fruit | TIME LEFT: 16.0s | Word: _ _ a _ | Guess:
The user attempts to guess a letter they have already used. The system displays an "Already guessed!" warning and prevents any deduction of score or time.
Winner
CONGRATULATIONS! YOU WON!
THE WORD WAS: BERRY
YOUR FINAL SCORE: 80
All underscores are replaced with correct letters. The game loop terminates successfully, displaying a "CONGRATULATIONS!" message and the final high score.
Game Over/Timeout
GAME OVER! Better luck next time.
THE WORD WAS: FROG
YOUR FINAL SCORE: 0
The time bank reaches 0.0s. The background timer triggers a "GAME OVER" state, breaks the loop, and reveals the secret word to the player.

Data Flow Diagram:



Data Dictionary:

Variable Name	Data Type	Format for display	Size in bytes	Size for display	Description	Example	Validation
word_bank	Dictionary	Internal	-900	N/A	It stores all the available words in the difficulty categories “1”, “2”, “3’ as keys for the system to use	{"1": ["apple"...]}	Type check: Dict. Keys must string "1"-"3".
categories	Dictionary	Internal	-1800	N/A	A table that matches each secret word into its connected themes.(e.g. Nature, Action)	{"wolf": "Animal"}	Key existence must match secret_word.
choice	String	N	50	1 char	Captures the user initial response to determine the difficulty of the game	“3”	if/else logic defaults invalid input to "1"
score	Integer	NNN	28	3 char	Tracks the player’s score and the score start differently, when difficulty is changed	80	Range check: value must be > 0 to continue.
secret_word	String	XXXXXXXX	58	4-12 char	A specific word that is selected in the word_bank (can be differed if user changes difficulty)	“algorithm”	Must be a valid string from word_bank.
display_word	List	X, X, X...	140+	10-24 char	a list of (dashes/strings) that are used to unveil the secret_word and visually track the	[‘a’ , ’_’ , ’_’]	len(display_word) must equal len(secret_word).

					player score and progress		
guessed_letters	List	X, X, X...	80+	Variable	Stores all the previous guessed letters so the player can't use the same letters otherwise being penalized	[‘p’ , ‘y’]	Membership check: if user_input is not in.
time_bank	Float	NN. N	24	4 char	This tracks the game time limit and is the reason the player gets 2+ more time every correct guess	14.2	Game terminates if time_bank <= 0.
start_loop	Float	Hidden	24	N/A	Records the exact time the game starts so it knows when the game ends when timer reaches 0	171041.....	Precise system time
current_input	String	X.X....	54	1-10 char	This will hold the letters you type for your guess and go when u press enters into the guessed_letters	“ki”	Accepts alphanumeric chars and \x08 (backspace)
msvcrt	Module	Not Available	N/A	N/A	This will let the game’s timer keep running even while the player is guessing the secret_word	Import msvcrt	Required for real-time game loop functionality.

Flowchart:

Flowchart for Hangman

This is the link to my Flowchart (had to be done because the flowchart quality was too low and unreadable when pasting onto my document)

Logbook For Hangman Game (Python)

Date	Time	Issue/Bugs	Solutions	Completed Task	What I Learnt	Bug Book
Feb 6 2026	1.4hrs	The starter code that was given was so considerably basic that it bored me in a few attempts and I needed to use tutorials to make it better but ran into code reading issues and the computer could not understand	I reviewed my code and tried reviewing and reviewing the code repeatedly was able to help find small issues in the code	I was able to find many issues and learn how to code a bit of python	The importance of consistent code formatting became clear during this session. Python is far more sensitive to whitespace than other programming languages, and debugging requires patience and systematic thinking rather than rushing to find quick fixes.	Function code blocks weren't indented properly (4 spaces), causing Python to treat them as separate code outside the function scope.
Feb 10, 2026	1.4hrs	While editing my earlier code I ran into almost 15 issues and had to restart some of the code due to bugs as they stopped the	I was able to find the issues that stopped the code from even appearing in terminal and	I was able to learn how to do edit python and I was able to create a title for hangman, and I was able to	Import order matters significantly in Python dependencies must be initialized	pygame.display and other pygame functions were called before, which must be executed first to

		code from even working in the terminal	searched up solutions to fix bugs	mess around with pygame to get a display to make hangman a bit more intriguing	before use. Choosing one graphics library prevents conflicting functionality, and clean code sometimes means removing features, not just adding them.	initialize all pygame modules.
Feb 11, 2026	50 mins P1	While trying to add a feature to give the player a chance to keep playing or not but it kept saying errors and it, I had to trial and error	I was able to get help from a video for tips on finding errors and faulty code that we can accidently leave behind	I was able to mess around with pygame a bit more and I tried integrating a display into my hangman game and for my default code I was able to add a restart feature	Exception handling using try-except blocks is essential for any program that accepts user input. Type conversion should never be assumed to work, it must be wrapped in error handling, and this principle has become a standard practice in all subsequent code.	The starter code's turtle graphics library conflicted with pygame, causing both libraries to fight for display window control and resulting in 15+ continuing errors.
Feb 11, 2026	50 mins P4	While trying to add more words to the word bank so there are more challenges and diverse options I ran into many errors as the computer could not read what I was asking it what to do	I was able to successfully integrate more words and more themes for these words, so the player feels much more engaged	I finished a word bank that has diverse themes and I started making an Easy/Medium/Hard difficult choice for people who want a challenge when it comes to hangman	Data structure integrity is critical—when multiple dictionaries reference each other, they must remain perfectly synchronized, or the program will crash at runtime.	The word_bank and categories dictionaries weren't synchronized—words were added to one but not the other, causing KeyError when the code tried to

					Data validation before access is a defensive programming practice that prevents errors from continuing through the system.	access missing keys.
Feb 16, 2026	50 mins P2	While starting my storyboard and flowchart it was very time consuming that I had to organize each aspect of my code	I was able to organize myself and make a draft for my storyboard and a learn how to use different features for better report	I was able to plan, and I was able to make a storyboard for my hangman game report	Planning is Necessary, the first draft doesn't need to be perfect; it's a starting point for refinement. Thinking through the game's structure from a user's perspective helps identify areas where code could be clearer or more robust.	The word_bank and categories dictionaries weren't synchronized words were added to one but not the other, causing KeyError when the code tried to access missing keys.
Feb 17, 2026	50 mins P2	While learning the different symbols of flowcharts I had a problem finding an application to help me make my flowcharts due to ads or subscriptions making it harder for me	I was able to find a flowchart designer app with no ads while it being free, and I was able to decide the difference between each of the symbols in a flowchart	I started my flowchart and made a dot-point plan associated to all the add-ons and the game features that are include in the game	Professional tools are often available for free if you search thoroughly. Understanding standards is essential for creating professional-quality documentation.	Initial attempts to use Lucid chart and Smart Draw resulted in excessive advertising and unstable software that made creating flowcharts difficult.

Feb 20, 2026	1.4hrs	While trying how to learn to use software is for flowcharts and DFD (data flow diagrams) every software had many bugs and did not save my progress, so it slowed my progression down a lot	I found some software like Lucid and Smart Draw, and they helped me make my DFD and flowcharts much more easily	I was able to start creating my flow chart and DFD for my portfolio	Cloud-based tools can be unreliable for time-sensitive project work, and local saves provide better security. Diagrams.net proved most stable for this project and became my preferred tool for all future diagramming work.	Lucidchart's cloud synchronization failed multiple times and SmartDraw crashed repeatedly, causing loss of unsaved progress on DFD and flowchart work.
Feb 23, 2026	1.4hrs	Today I was able to focus on my code, and I was able to make a foolproof plan that can help me make more added features and the smallest features that can make the hangman game more intriguing	I added more hangman related features that make the game fun and challenging at the same time and I was able to fix some issues while adding the features	I started making more features and planned for my added features for a timer in my code and I am watching videos on how to apply it	Professional software development prioritizes planning before implementation. Research and planning phases save significant time during actual coding.	First approach to timer implementation used time.sleep(), which was later discovered to block the game loop and prevent input processing.
Mar 2, 2026	50min	Today I was finalizing my code and added the final additional feature which was the timer for my hangman code which intensifies the game and makes the game more fun but the timer had many problems on how it worked on python and how to apply it to my code as it gave problems to my inputs	I was able to fix the problems that came across adding a timer and I was able to get a fully working timer that refresh every 0.1s to update the timer acting like an actual timer	I was able to get my timer working and I am incredibly happy with my code progress because I have finished my finalized hangman code	Blocking versus non-blocking input/output is critical for game development. should be used to track elapsed time as a delta from a starting point, not as an absolute value. Non-blocking approaches are	Recalculating the start_time variable inside the game loop instead of using a fixed reference point caused elapsed time calculations to be incorrect after multiple guesses.

					essential for maintaining game loop responsiveness.	
Mar 6, 2026	1.4hrs	Today I edited my code, and I edited my storyboard for a simpler but effective design and added descriptions to each storyboard panel which gave it more depth	One problem that stuck out the most was the fact that the quality went down every time I screen clip the necessary images, I need to help the viewer visualize the code's function	Finished Storyboard and Finished the Finalized Code	Visual documentation must be accompanied by clear text explanations—images alone are insufficient for effective communication. Screenshot quality impacts professionalism of documentation.	Screenshots taken with default resolution settings resulted in pixelated, low-contrast images that made UI elements difficult to distinguish.
Mar 9. 2026	1.4hrs	I finished my Data Dictionary and Started to make My Data Flow Diagram (DFD) but while doing my data dictionary I only had trouble trying to remember where I put my variables in my code and I am trying to learn how to make a DFD so I can get a full 5 marks	So, I used I read my whole Code once which was not that effective, but it still helped with finding all my codes and it was like a win-win where I could revise, to learn my code	Finished Data Dictionary, Started my DFD	Detailed code documentation is easier when code is well-organized. Writing documentation helps identify missing or poorly designed variables. Thorough code review deepens understanding of your own work and reveals code quality and structure.	Local variables defined inside one function cannot be accessed from other functions without proper parameter passing, causing potential Name Error exceptions.

Mar 15, 2026	1.5hrs	While learning how to make an Extensive Flowchart I ran into problems while getting the right shapes for the right input and output and I had to learn how a flowchart works to get a good understanding	So, i watched YouTube videos to help me learn the best software and websites to make a flowchart and the correct shapes for every single input and output	Finished the Flowchart and Improved the Data Dictionary	Flowcharts should be high-level but detailed enough to code from. Correct symbols of matter for professional communication. Reviewing standards of documents and video resources ensures that documentation meets professional expectations.	Initial flowchart used incorrect symbols for operations (rectangles for decisions, diamonds for processes), violating NESA standards, and confusing readers.
Mar 24th, 2026	1.5hrs	Today, while trying to refine my document, I found numerous problems trying to fix the graphs and data in my tables, and I found it hard to make a small and concise flowchart while maintaining the same inputs and outputs	I was able to Discuss with my teacher (Mr. Wright) about how to solve this problem, we were able to agree on giving him a link on the document itself to the flowchart i created and changing the format of my report so i can fit all my data	Finalized Flowchart, Improved Logbook even more and Refined my Report	Refinement is an important part of the documentation process. Clear labeling and consistent formatting make documentation more professional and easier to understand. Adding tracking columns to development logs helps identify patterns in the development process and provides better documentation of the journey.	Some decision branches in the flowchart could be consolidated without losing important logic, making the diagram unnecessarily complex.

Reflection On the Report:

AI Usage Example:

I used a software called Claude Code, I used this software to debug a deduction bug on my hangman code, I asked it to identify why and where this bug was. I carefully read the explanation that was given to me and rather accepting newly generated code, I found the bug with its help and figured out that I put the deduction logic in two if statements, thus resulting in the score-loss doubling and this mistake was because of me not clearly looking at my code properly. Claude Code was able to help me because while explaining the problem, it also explained the concept of time.time(), which then I used to improve and refine my code which was much more effective. However, this could've all been avoided if I had planned and organized my plan for the timer being added to my code.

Significant Challenges & Solutions:

Timer Blocking: This was one of the most tedious and trickiest parts of my code. I didn't know how to make the timer like a live countdown and a normal input() would just pause the timer until user guessed and show new timer after the user's guess which wasn't what I was aiming for. For my solution I went to YouTube tutorials, and many were long and unreliable and many I couldn't understand, so it took me a while to grasp the concept, but after many YouTube tutorials I was able to fix the problem and get my timer to do live countdown. This all could've been avoided, if I had found more faster and effective ways to teach myself hard concepts.

Dictionary Key Matches: This was a very tedious problem, and this made me frustrated while I was adding words to word_bank but forgot to add matching entries in categories, causing Key Error crashes. My solution was systematically verifying every word had a corresponding category and adding validation checks. The solution worked but revealed I wasn't thinking about data relationships before coding.

Learning about DFDs: At first, every time I looked at a DFD, it looked way awfully similar to a Flowchart, and it confused me on how a DFD should look like and how I should make one, So I had to do some research and understand that DFD show the data process in a system. For my solution, I studied examples and created my DFD step-by-step, mapping player input flowing to game logic, then to score updates. This process was effective because it forced me to think about the system structure from a data perspective, which revealed dependencies I hadn't considered before.

What I Learned About Programming, Planning & SDLC:

I learned that careful planning prevents debugging nightmares. Every bug in my logbook could have been caught earlier with better thinking before coding. I grew from rushing into code to understanding that architecture and data design matter more than typing speed. My debugging process taught me that understanding why code fails is more valuable than just fixing it. It prevents the same mistake from happening again.

Bibliography:

GeeksforGeeks. (n.d.). How to create a countdown timer using Python. https://www.geeksforgeeks.org/python/how-to-create-a-countdown-timer-using-python/

Grok Learning. (2024). The Grok Academy. https://groklearning.com/

O'Reilly Media. (2003). Python standard library reference: msvcrt - Windows-specific API. https://www.oreilly.com/library/view/python-standard-library/0596000960/ch12s11.html

Tech With Tim. (2022). Python hangman game tutorial [Video]. YouTube.
https://www.youtube.com/watch?v=ag8NtD1e0Kc

YouTube Creator. (2022). Python game development tutorial [Video]. YouTube.
https://www.youtube.com/watch?v=m4nEnsavl6w

YouTube Creator. (2022). Python input handling and user interaction [Video]. YouTube.
https://www.youtube.com/watch?v=kqtD5dpn9C8

YouTube Creator. (2022). Python data structures and dictionaries [Video]. YouTube.
https://www.youtube.com/watch?v=Sg4GMVMdOPo&list=PLZPZq0r_RZOOkUQbat8LyQii36cJf2SWT

YouTube Creator. (2022). Python non-blocking input with msvcrt [Video]. YouTube.
https://www.youtube.com/watch?v=kUt0nS0yMtM

YouTube Creator. (2022). Python timer and time module tutorial [Video]. YouTube.
https://www.youtube.com/watch?v=6VGTvgaJllM

YouTube Creator. (2022). Python debugging and error handling [Video]. YouTube.
https://www.youtube.com/watch?v=KseiSR0MCTI